Lesson 29 material: Expressing Your Results and Conducting Your Analyses

12.3: Expressing Your Results

From *Research Methods in Psychology (4th American edition)* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0). This section: CC BY-NC-SA 4.0; authored, remixed or curated by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton via source content that was edited to the style and standards of the LibreTexts platform.
Original page, with audio and online exercises:
https://socialsci.libretexts.org/Bookshelves/Psychology/Research_Methods_and_Statistics/Research_Methods_in_Psychology_(Jhangiani_Chiang_Cuttler_and_Leighton)/12%3
Changes: formatted for print; exercise answers are in the study guide.

Learning Objectives

1. Write out simple descriptive statistics in American Psychological Association (APA) style.
2. Interpret and create simple APA-style figures—including bar graphs, line graphs, and scatterplots.
3. Interpret and create simple APA-style tables—including tables of group or condition means and correlation matrices.

Once you have conducted your descriptive statistical analyses, you will need to present them to others. In this section, we focus on presenting descriptive statistical results in writing, in figures, and in tables—following American Psychological Association (APA) guidelines for written research reports. These principles can be adapted easily to other presentation formats such as posters and slide show presentations.

Presenting Descriptive Statistics in Writing

Recall that APA style includes several rules for presenting numerical results in the text (see 4.31–4.34 in the APA *Publication Manual*) . These include using words only for numbers less than 10 that do not represent precise statistical results and using numerals for numbers 10 and higher. However, statistical results are always presented in the form of numerals rather than words and are usually rounded to two decimal places (e.g., “2.00” rather than “two” or “2”). They can be presented either in the narrative description of the results or parenthetically—much like reference citations. When you have a small number of results to report, it is often most efficient to write them out. Here are some examples:

The mean age of the participants was 22.43 years with a standard deviation of 2.34.

Among the participants with low self-esteem, those in a negative mood expressed stronger intentions to have unprotected sex (M = 4.05, SD = 2.32) than those in a positive mood (M = 2.15, SD = 2.27).

The treatment group had a mean of 23.40 (SD = 9.33), while the control group had a mean of 20.87 (SD = 8.45).

The test-retest correlation was .96.

There was a moderate negative correlation between the alphabetical position of respondents’ last names and their response time (r = –.27).

Notice that when presented in the narrative, the terms *mean* and *standard deviation* are written out, but when presented parenthetically, the symbols M and SD are used instead. Notice also that it is especially important to use parallel construction to express similar or comparable results in similar ways. The third example is *much* better than the following nonparallel alternative:

The treatment group had a mean of 23.40 (SD = 9.33), while 20.87 was the mean of the control group, which had a standard deviation of 8.45.

Presenting Descriptive Statistics in Figures

When you have a large number of results to report, you can often do it more clearly and efficiently with a graphical depiction of the data, such as pie charts, bar graphs, or scatterplots. In an APA style research report, these graphs are presented as **figures**. When you prepare figures for an APA-style research report, there are some general guidelines that you should keep in mind. First, the figure should always add important information rather than repeat information that already appears in the text or in a table (if a figure presents information more clearly or efficiently, then you should keep the figure and eliminate the text or table.) Second, figures should be as simple as possible. For example, the *Publication Manual* discourages the use of color unless it is absolutely necessary (although color can still be an effective element in posters, slide show presentations, or textbooks.) Third, figures should be interpretable on their own. A reader should be able to understand the basic result based only on the figure and its caption and should not have to refer to the text for an explanation.

There are also several more technical guidelines for presentation of figures that include the following (see the APA Publication Manual section 5.20 through 5.30):

- Layout of graphs
  - In general, scatterplots, bar graphs, and line graphs should be slightly wider than they are tall.
  - The independent variable should be plotted on the x- axis and the dependent variable on the y- axis.
  - Values should increase from left to right on the x- axis and from bottom to top on the y- axis.
  - The x-axis and y-axis should begin with the value zero.
- Axis Labels and Legends
  - Axis labels should be clear and concise and include the units of measurement if they do not appear in the caption.
  - Axis labels should be parallel to the axis.
  - Legends should appear within the figure.
  - Text should be in the same simple font throughout and no smaller than 8 point and no larger than 14 point.
- Captions

- Captions are titled with the word “Figure”, followed by the figure number in the order in which it appears in the text, and terminated with a period. This title is italicized.
- After the title is a brief description of the figure terminated with a period (e.g., “Reaction times of the control versus experimental group.”)
- Following the description, include any information needed to interpret the figure, such as any abbreviations, units of measurement (if not in the axis label), units of error bars, etc.

“Convincing” retrieved from http://imgs.xkcd.com/comics/convincing.png (CC-BY-NC 2.5)

Bar Graphs

As we have seen throughout this book, **bar graphs** are generally used to present and compare the mean scores for two or more groups or conditions. The bar graph in Figure is an APA-style version of Figure 12.4. Notice that it conforms to all the guidelines listed. A new element in Figure is the smaller vertical bars that extend both upward and downward from the top of each main bar. These are **error bars**, and they represent the variability in each group or condition. Although they sometimes extend one standard *deviation* in each direction, they are more likely to extend one standard *error* in each direction (as in Figure). The **standard error** is the standard deviation of the group divided by the square root of the sample size of the group. The standard error is used because, in general, a difference between group means that is greater than two standard errors is statistically significant. Thus one can “see” whether a difference is statistically significant based on a bar graph with error bars.

Figure: Mean clinician's rating of phobia severity for participants receiving the education treatment and the exposure treatment. Error bars represent standard errors. Sample APA-Style Bar Graph, With Error Bars Representing the Standard Errors, Based on Research by Ollendick and Colleagues

Line Graphs

Line graphs are used when the independent variable is measured in a more continuous manner (e.g., time) or to present correlations between quantitative variables when the independent variable has, or is organized into, a relatively small number of distinct levels. Each point in a line graph represents the mean score on the dependent variable for participants at one level of the independent variable. Figure is an APA-style version of the results of Carlson and Conard. Notice that it includes error bars representing the standard error and conforms to all the stated guidelines.

Figure: Mean response time by the alphabetical position of respondents' names in the alphabet. Response times are expressed as z scores. Error bars represent standard errors. Sample APA-Style Line Graph Based on Research by Carlson and Conard

In most cases, the information in a line graph could just as easily be presented in a bar graph. In Figure , for example, one could replace each point with a bar that reaches up to the same level and leave the error bars right where they are. This emphasizes the fundamental similarity of the two types of statistical relationship. Both are differences in the average score on one variable across levels of another. The convention followed by most researchers, however, is to use a bar graph when the variable plotted on the *x*- axis is categorical and a line graph when it is quantitative.

Scatterplots

Scatterplots are used to present correlations and relationships between quantitative variables when the variable on the *x*- axis (typically the independent variable) has a large number of levels. Each point in a scatterplot represents an individual rather than the mean for a group of individuals, and there are no lines connecting the points. The graph in Figure is an APA-style version of Figure 12.7, which illustrates a few additional points. First, when the variables on the *x*- axis and *y*-axis are conceptually similar and measured on the same scale—as here, where they are measures of the same variable on two different occasions—this can be emphasized by making the axes the same length. Second, when two or more individuals fall at exactly the same point on the graph, one way this can be indicated is by offsetting the points slightly along the *x*- axis. Other ways are by displaying the number of individuals in parentheses next to the point or by making the point larger or darker in proportion to the number of individuals. Finally, the straight line that best fits the points in the scatterplot, which is called the regression line, can also be included.

Figure: Sample APA-Style Scatterplot. Relationship between scores on the Rosenberg self-esteem scale taken by 25 research methods students on two occasions one week apart. Pearson's r=0.96.

Expressing Descriptive Statistics in Tables

Like graphs, tables can be used to present large amounts of information clearly and efficiently. The same general principles apply to tables as apply to graphs. They should add important information to the presentation of your results, be as simple as possible, and be interpretable on their own. Again, we focus here on tables for an APA-style manuscript.

The most common use of tables is to present several means and standard deviations—usually for complex research designs with multiple independent and dependent variables. Figure , for example, shows the results of a hypothetical study similar to the one by MacDonald and Martineau (2002)[1] (The means in Figure are the means reported by MacDonald and Martineau, but the standard errors are not). Recall that these researchers categorized participants as having low or high self-esteem, put them into a negative or positive mood, and measured their intentions to have unprotected sex. They also measured participants’ attitudes toward unprotected sex. Notice that the table includes horizontal lines spanning the entire table at the top and bottom, and just beneath the column headings. Furthermore, every column has a heading—including the leftmost column—and there are additional headings that span two or more columns that help to organize the information and present it more efficiently. Finally, notice that APA-style tables are numbered consecutively starting at 1 and given a brief but clear and descriptive title.

Figure: Sample APA-Style Table Presenting Means and Standard Deviations. Means and Standard Deviations of Intentions to Have Unprotected Sex and Attitudes Toward Unprotected Sex as a Function of Both Mood and Self-Esteem.

Another common use of tables is to present correlations—usually measured by Pearson’s *r*—among several variables. This kind of table is called a **correlation matrix**. Figure is a correlation matrix based on a study by David McCabe and colleagues (McCabe, Roediger, McDaniel, Balota, & Hambrick, 2010)[2]. They were interested in the relationships between working memory and several other variables. We can see from the table that the correlation between working memory and executive function, for example, was an extremely strong .96, that the correlation between working memory and vocabulary was a medium 0.27, and that all the measures except vocabulary tend to decline with age. Notice here that only half the table is filled in because the other half would have identical values. For example, the Pearson’s *r* value in the upper right corner (working memory and age) would be the

same as the one in the lower left corner (age and working memory). The correlation of a variable with itself is always 1.00, so these values are replaced by dashes to make the table easier to read.

Figure: Sample APA-Style Table (Correlation Matrix) Based on Research by McCabe and Colleagues. Correlations Between Five Cognitive Variables and Age.

As with graphs, precise statistical results that appear in a table do not need to be repeated in the text. Instead, the writer can note major trends and alert the reader to details (e.g., specific correlations) that are of particular interest.

References

1. MacDonald, T. K., & Martineau, A. M. (2002). Self-esteem, mood, and intentions to use condoms: When does low self-esteem lead to risky health behaviors? *Journal of Experimental Social Psychology, 38*, 299–306. ↵
2. McCabe, D. P., Roediger, H. L., McDaniel, M. A., Balota, D. A., & Hambrick, D. Z. (2010). The relationship between working memory capacity and executive functioning. *Neuropsychology, 24*(2), 222–243. doi:10.1037/a0017619 ↵

Links on the original page

- Web page (imgs.xkcd.com): http://imgs.xkcd.com/comics/convincing.png

12.4: Conducting Your Analyses

From *Research Methods in Psychology (4th American edition)* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0). This section: CC BY-NC-SA 4.0; authored, remixed or curated by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton via source content that was edited to the style and standards of the LibreTexts platform.
Original page, with audio and online exercises:
https://socialsci.libretexts.org/Bookshelves/Psychology/Research_Methods_and_Statistics/Research_Methods_in_Psychology_(Jhangiani_Chiang_Cuttler_and_Leighton)/12%
Changes: formatted for print; exercise answers are in the study guide.

Learning Objective

1. Describe the steps involved in preparing and analyzing a typical set of raw data.
2. Differentiate between planned and exploratory data analyses.

Even when you understand the statistics involved, analyzing data can be a complicated process. It is likely that for each of several participants, there are data for several different variables: demographics such as sex and age, one or more independent variables, one or more dependent variables, and perhaps a manipulation check. Furthermore, the "raw" (unanalyzed) data might take several different forms—completed paper-and-pencil questionnaires, computer files filled with numbers or text, videos, or written notes—and these may have to be organized, coded, or combined in some way. There might even be missing, incorrect, or just "suspicious" responses that must be dealt with. In this section, we consider some practical advice to make this process as organized and efficient as possible.

Prepare Your Data for Analysis

Whether your raw data are on paper or in a computer file (or both), there are a few things you should do before you begin analyzing them. First, be sure they do not include any information that might identify individual participants and be sure that you have a secure location where you can store the data and a separate secure location where you can store any consent forms. Unless the data are highly sensitive, a locked room or password-protected computer is usually good enough. It is also a good idea to make photocopies or backup files of your data and store them in yet another secure location—at least until the project is complete. Professional researchers usually keep a copy of their raw data and consent forms for several years in case questions about the procedure, the data, or participant consent arise after the project is completed.

Next, you should check your **raw data** to make sure that they are complete and appear to have been accurately recorded (whether it was participants, yourself, or a computer program that did the recording). At this point, you might find that there are illegible or missing responses, or obvious misunderstandings (e.g., a response of "12" on a 1-to-10 rating scale). You will have to decide whether such problems are severe enough to make a participant's data unusable. If information about the main independent or dependent variable is missing, or if several responses are missing or suspicious, you may have to exclude that participant's data from the analyses. If you do decide to exclude any data, do not throw them away or delete them because you or another researcher might want to see them later. Instead, set them aside and keep notes about why you decided to exclude them because you will need to report this information.

Now you are ready to enter your data in a spreadsheet program or, if it is already in a computer file, to format it for analysis. You can use a general spreadsheet program like Microsoft Excel or a statistical analysis program like SPSS to create your **data file**. (Data files created in one program can usually be converted to work with other programs.) The most common format is for each row to represent a participant and for each column to represent a variable (with the variable name at the top of each column). A sample data file is shown in Table . The first column contains participant identification numbers. This is followed by columns containing demographic information (sex and age), independent variables (mood, four self-esteem items, and the total of the four self-esteem items), and finally dependent variables (intentions and attitudes). Categorical variables can usually be entered as category labels (e.g., "M" and "F" for male and female) or as numbers (e.g., "0" for negative mood and "1" for positive mood). Although category labels are often clearer, some analyses might require numbers. SPSS allows you to enter numbers but also attach a category label to each number.

Table : Sample Data File

ID	SEX	AGE	MOOD	SE1	SE2	SE3	SE4	TOTAL	INT	ATT
1	M	20	1	2	3	2	3	10	6	5
2	F	22	1	1	0	2	1	4	4	4
3	F	19	0	2	2	2	2	8	2	3
4	F	24	0	3	3	2	3	11	5	6

If you have multiple-response measures—such as the self-esteem measure in Table —you could combine the items by hand and then enter the total score in your spreadsheet. However, it is much better to enter each response as a separate variable in the spreadsheet—as with the self-esteem measure in Table —and use the software to combine them (e.g., using the "AVERAGE" function in Excel or the "Compute" function in SPSS). Not only is this approach more accurate, but it allows you to detect and correct errors, to assess internal consistency, and to analyze individual responses if you decide to do so later.

Preliminary Analyses

Before turning to your primary research questions, there are often several preliminary analyses to conduct. For multiple-response measures, you should assess the internal consistency of the measure. Statistical programs like SPSS will allow you to compute Cronbach's α or Cohen's κ. If this is beyond your comfort level, you can still compute and evaluate a split-half correlation.

Next, you should analyze each important variable separately. (This step is not necessary for manipulated independent variables, of course, because you as the researcher determined what the distribution would be.) Make histograms for each one, note their shapes, and compute the common measures of central tendency and variability. Be sure you understand what these statistics *mean* in terms of the variables you are interested in. For example, a distribution of self-report happiness ratings on a 1-to-10-point scale might be unimodal and negatively skewed with a mean of 8.25 and a standard

deviation of 1.14. But what this *means* is that most participants rated themselves fairly high on the happiness scale, with a small number rating themselves noticeably lower.

Now is the time to identify outliers, examine them more closely, and decide what to do about them. You might discover that what at first appears to be an outlier is the result of a response being entered incorrectly in the data file, in which case you only need to correct the data file and move on. Alternatively, you might suspect that an outlier represents some other kind of error, misunderstanding, or lack of effort by a participant. For example, in a reaction time distribution in which most participants took only a few seconds to respond, a participant who took 3 minutes to respond would be an outlier. It seems likely that this participant did not understand the task (or at least was not paying very close attention). Also, including their reaction time would have a large impact on the mean and standard deviation for the sample. In situations like this, it can be justifiable to exclude the outlying response or participant from the analyses. If you do this, however, you should keep notes on which responses or participants you have excluded and why, and apply those same criteria consistently to every response and every participant. When you present your results, you should indicate how many responses or participants you excluded and the specific criteria that you used. And again, do not literally throw away or delete the data that you choose to exclude. Just set them aside because you or another researcher might want to see them later.

Keep in mind that outliers do not *necessarily* represent an error, misunderstanding, or lack of effort. They might represent truly extreme responses or participants. For example, in one large university student sample, the vast majority of participants reported having had fewer than 15 sexual partners, but there were also a few extreme scores of 60 or 70 (Brown & Sinclair, 1999)[1]. Although these scores might represent errors, misunderstandings, or even intentional exaggerations, it is also plausible that they represent honest and even accurate estimates. One strategy here would be to use the median and other statistics that are not strongly affected by the outliers. Another would be to analyze the data both including and excluding any outliers. If the results are essentially the same, which they often are, then it makes sense to leave the outliers. If the results differ depending on whether the outliers are included or excluded them, then both analyses can be reported and the differences between them discussed.

Planned and Exploratory Analyses

Finally, you are ready to answer your primary research questions. When you designed your study, you might have had a hypothesis that a particular relationship might exist in the data. In this case, you would conduct a **planned analysis**, to test a relationship that you expected in your hypothesis. For example, if you expected a difference between group or condition means, you can compute the relevant group or condition means and standard deviations, make a bar graph to display the results, and compute Cohen's *d*. If you expected a correlation between quantitative variables, you can make a line graph or scatterplot (be sure to check for nonlinearity and restriction of range) and compute Pearson's *r*.

Once you have conducted your planned analyses, you can move on to examine the possibility there might be relationships in the data that you did not hypothesize. This would be an **exploratory analysis**, an analysis that you are undertaking without an existing hypothesis. These analyses will help you explore your data for other interesting results that might provide the basis for future research (and material for the discussion section of your paper). Daryl Bem (2003) suggests that you

[e]xamine [your data] from every angle. Analyze the sexes separately. Make up new composite indexes. If a datum suggests a new hypothesis, try to find additional evidence for it elsewhere in the data. If you see dim traces of interesting patterns, try to reorganize the data to bring them into bolder relief. If there are participants you don't like, or trials, observers, or interviewers who gave you anomalous results, drop them (temporarily). Go on a fishing expedition for something—anything—interesting. (p. 186–187)[2]

It is important to differentiate planned from exploratory analyses in writing your results and discussion sections of your report. This is because complex sets of data are likely to include "patterns" that occurred entirely by chance, and every time you do another unplanned analysis on these data, you increase the likelihood these chance patterns will appear to be real patterns, what is referred to as a "Type 1" error (see the chapter on Inferential Statistics). Thus results discovered while doing exploratory analyses (what Bem calls a "fishing expedition") should be viewed skeptically and replicated in at least one new study before being presented. But, if you do find interesting relationships you did not expect in the data, explain that they might be worthy of additional research.

Understand Your Descriptive Statistics

In the next chapter, we will consider inferential statistics—a set of techniques for deciding whether the results for your sample are likely to apply to the population. Although inferential statistics are important for reasons that will be explained shortly, beginning researchers sometimes forget that their descriptive statistics really tell "what happened" in their study. For example, imagine that a treatment group of 50 participants has a mean score of 34.32 (SD = 10.45), a control group of 50 participants has a mean score of 21.45 (SD = 9.22), and Cohen's *d* is an extremely strong 1.31. Although conducting and reporting inferential statistics (like a *t* test) would certainly be a required part of any formal report on this study, it should be clear from the descriptive statistics alone that the treatment worked. Or imagine that a scatterplot shows an indistinct "cloud" of points and Pearson's *r* is a trivial −.02. Again, although conducting and reporting inferential statistics would be a required part of any formal report on this study, it should be clear from the descriptive statistics alone that the variables are essentially unrelated. The point is that you should always be sure that you thoroughly understand your results at a descriptive level first, and then move on to the inferential statistics.

References

1. Brown, N. R., & Sinclair, R. C. (1999). Estimating number of lifetime sexual partners: Men and women do it differently. *The Journal of Sex Research, 36*, 292–297. ↵
2. Bem, D. J. (2003). Writing the empirical journal article. In J. M. Darley, M. P. Zanna, & H. L. Roediger III (Eds.), *The complete academic: A career guide* (2nd ed., pp. 185–219). Washington, DC: American Psychological Association. ↵

12.5: Descriptive Statistics (Summary)

From *Research Methods in Psychology (4th American edition)* by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0). This section: CC BY-NC-SA 4.0; authored, remixed or curated by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler, and Dana C. Leighton via source content that was edited to the style and standards of the LibreTexts platform.
Original page, with audio and online exercises:
https://socialsci.libretexts.org/Bookshelves/Psychology/Research_Methods_and_Statistics/Research_Methods_in_Psychology_(Jhangiani_Chiang_Cuttler_and_Leighton)/12%
Changes: formatted for print; exercise answers are in the study guide.

Key Takeaways

- Every variable has a distribution—a way that the scores are distributed across the levels. The distribution can be described using a frequency table and histogram. It can also be described in words in terms of its shape, including whether it is unimodal or bimodal, and whether it is symmetrical or skewed.
- The central tendency, or middle, of a distribution can be described precisely using three statistics—the mean, median, and mode. The mean is the sum of the scores divided by the number of scores, the median is the middle score, and the mode is the most common score.
- The variability, or spread, of a distribution can be described precisely using the range and standard deviation. The range is the difference between the highest and lowest scores, and the standard deviation is the average amount by which the scores differ from the mean.
- The location of a score within its distribution can be described using percentile ranks or *z* scores. The percentile rank of a score is the percentage of scores below that score, and the *z* score is the difference between the score and the mean divided by the standard deviation.
- Differences between groups or conditions are typically described in terms of the means and standard deviations of the groups or conditions or in terms of Cohen's *d* and are presented in bar graphs.
- Cohen's *d* is a measure of relationship strength (or effect size) for differences between two group or condition means. It is the difference of the means divided by the standard deviation. In general, values of ±0.20, ±0.50, and ±0.80 can be considered small, medium, and large, respectively.
- Correlations between quantitative variables are typically described in terms of Pearson's *r* and presented in line graphs or scatterplots.
- Pearson's *r* is a measure of relationship strength (or effect size) for relationships between quantitative variables. It is the mean cross-product of the two sets of *z* scores. In general, values of ±.10, ±.30, and ±.50 can be considered small, medium, and large, respectively.
- In an APA-style article, simple results are most efficiently presented in the text, while more complex results are most efficiently presented in graphs or tables.
- APA style includes several rules for presenting numerical results in the text. These include using words only for numbers less than 10 that do not represent precise statistical results, and rounding results to two decimal places, using words (e.g., "mean") in the text and symbols (e.g., "*M*") in parentheses.
- APA style includes several rules for presenting results in graphs and tables. Graphs and tables should add information rather than repeating information, be as simple as possible, and be interpretable on their own with a descriptive caption (for graphs) or a descriptive title (for tables).
- Raw data must be prepared for analysis by examining them for possible errors, organizing them, and entering them into a spreadsheet program.
- Preliminary analyses on any data set include checking the reliability of measures, evaluating the effectiveness of any manipulations, examining the distributions of individual variables, and identifying outliers.
- Outliers that appear to be the result of an error, a misunderstanding, or a lack of effort can be excluded from the analyses. The criteria for excluded responses or participants should be applied in the same way to all the data and described when you present your results. Excluded data should be set aside rather than destroyed or deleted in case they are needed later.
- Descriptive statistics tell the story of what happened in a study. Although inferential statistics are also important, it is essential to understand the descriptive statistics first.

References

Ollendick, T. H., Öst, L.-G., Reuterskiöld, L., Costa, N., Cederlund, R., Sirbu, C.,...Jarrett, M. A. (2009). One-session treatments of specific phobias in youth: A randomized clinical trial in the United States and Sweden. *Journal of Consulting and Clinical Psychology, 77*, 504–516.

Cohen, J. (1992). A power primer. *Psychological Bulletin, 112*, 155–159.

Hyde, J. S. (2007). New directions in the study of gender similarities and differences. *Current Directions in Psychological Science, 16*, 259–263.

Carlson, K. A., & Conard, J. M. (2011). The last name effect: How last name influences acquisition timing. *Journal of Consumer Research, 38*(2), 300-307. doi: 10.1086/658470

MacDonald, T. K., & Martineau, A. M. (2002). *Self-esteem, mood, and intentions to use condoms: When does low self-esteem lead to risky health behaviors? Journal of Experimental Social Psychology, 38*, 299–306.

McCabe, D. P., Roediger, H. L., McDaniel, M. A., Balota, D. A., & Hambrick, D. Z. (2010). *The relationship between working memory capacity and executive functioning*. *Neuropsychology, 24*(2), 222–243. doi:10.1037/a0017619

Brown, N. R., & Sinclair, R. C. (1999). Estimating number of lifetime sexual partners: Men and women do it differently. *The Journal of Sex Research, 36,* 292–297.

Bem, D. J. (2003). Writing the empirical journal article. In J. M. Darley, M. P. Zanna, & H. L. Roediger III (Eds.), *The complete academic: A career guide* (2nd ed., pp. 185–219). Washington, DC: American Psychological Association.

Schmitt, D. P., & Allik, J. (2005). Simultaneous administration of the Rosenberg Self-Esteem Scale in 53 nations: Exploring the universal and culture-specific features of global self-esteem. *Journal of Personality and Social Psychology, 89*, 623–642.

Buss, D. M., & Schmitt, D. P. (1993). Sexual strategies theory: A contextual evolutionary analysis of human mating. *Psychological Review, 100*, 204–232.

Exercises

- Practice: Make a frequency table and histogram for the following data. Then write a short description of the shape of the distribution in words.
  - 11, 8, 9, 12, 9, 10, 12, 13, 11, 13, 12, 6, 10, 17, 13, 11, 12, 12, 14, 14
- Practice: For the data in Exercise 1, compute the mean, median, mode, standard deviation, and range.
- Practice: Using the data in Exercises 1 and 2, find
  - the percentile ranks for scores of 9 and 14
  - the *z* scores for scores of 8 and 12.
- Practice: The following data represent scores on the Rosenberg Self-Esteem Scale for a sample of 10 Japanese university students and 10 American university students. (Although hypothetical, these data are consistent with empirical findings [Schmitt & Allik, 2005][1].) Compute the means and standard deviations of the two groups, make a bar graph, compute Cohen's *d*, and describe the strength of the relationship in words.

Japan	**United States**
25	27
20	30
24	34
28	37
30	26
32	24
21	28
24	35
20	33
26	36

- Practice: The hypothetical data that follow are extraversion scores and the number of Facebook friends for 15 university students. Make a scatterplot for these data, compute Pearson's *r*, and describe the relationship in words.

Extraversion	**Facebook Friends**
8	75
10	315
4	28
6	214
12	176
14	95
10	120
11	150
4	32
13	250
5	99
7	136
8	185
11	88
10	144

- Practice: In a classic study, men and women rated the importance of physical attractiveness in both a short-term mate and a long-term mate (Buss & Schmitt, 1993)[2]. The means and standard deviations are as follows. Men / Short Term: $M = 5.67$, $SD = 2.34$; Men / Long Term: $M = 4.43$, $SD = 2.11$; Women / Short Term: $M = 5.67$, $SD = 2.48$; Women / Long Term: $M = 4.22$, $SD = 1.98$. Present these results
  - in writing
  - in a figure
  - in a table
- Discussion: What are at least two reasonable ways to deal with each of the following outliers based on the discussion in this chapter? (a) A participant estimating ordinary people's heights estimates one woman's height to be "84 inches" tall. (b) In a study of memory for ordinary objects, one participant scores 0 out of 15. (c) In response to a question about how many "close friends" she has, one participant writes "32."

References

1. Schmitt, D. P., & Allik, J. (2005). Simultaneous administration of the Rosenberg Self-Esteem Scale in 53 nations: Exploring the universal and culture-specific features of global self-esteem. *Journal of Personality and Social Psychology, 89*, 623–642. ↵
2. Buss, D. M., & Schmitt, D. P. (1993). Sexual strategies theory: A contextual evolutionary analysis of human mating. *Psychological Review, 100*, 204–232. ↵